



AI/ML IN SECURITY

CREDIT CARD FRAUD DETECTION SYSTEM

By -

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Problem Statement – The Financial Fraud Crisis

- **The Global Threat Landscape:** As digital transaction volumes hit record highs, traditional security measures are failing to keep pace with sophisticated "Synthetic Identity" and "Card-Not-Present" fraud.
- **Limitations of Legacy Systems:** Static rule-based engines (e.g., "Flag if amount > \$5000") are too predictable. Scammers easily bypass these thresholds by staying just below them.
- **The Cybersecurity Gap:** Modern fraud requires a "Pattern-First" approach. Without automated detection, banks face massive capital loss, legal liabilities, and a breakdown in consumer trust.
- **Legal Necessity:** Regulatory bodies (RBI/SEBI) now mandate "Reasonable Security Practices." Failing to detect preventable fraud can lead to heavy penalties under the IT Act.



Project Objectives & System Architecture

- **End-to-End Automation:** To develop a high-speed analytical pipeline that ingests raw transaction logs and outputs actionable risk scores.
- **ML-Powered Accuracy:** Utilizing Ensemble Learning (Random Forest) to detect non-linear fraud patterns that human auditors would miss.
- **The Architecture:**
 - **Frontend:** Minimalist dashboard for data upload and visualization.
 - **Backend (Python/Flask):** The central hub for file handling, data cleaning, and model routing.
 - **ML Layer:** A pre-trained classifier capable of millisecond inference.
- **Scalability:** Designed to handle thousands of transactions in a single batch processing request.

Exploratory Data Analysis (EDA) & Preprocessing

- **Feature Correlation:** Analyzing how transaction magnitude (Amount) and merchant categories correlate with fraudulent behavior.
- **Handling Class Imbalance:** Fraud is rare (often $<1\%$ of data). Our EDA process ensures the model doesn't become "biased" toward legitimate transactions.
- **The Data Pipeline:**
 - **Cleaning:** Removing corrupted entries and handling missing values via Pandas.
 - **Scaling:** Normalizing numerical features so high-value transactions don't skew the model's perception.
 - **Encoding:** Converting categorical metadata into mathematical vectors for model ingestion.

Machine Learning Engine: Random Forest Classifier

- **Ensemble Intelligence:** Random Forest operates by constructing a multitude of decision trees. It outputs the class that is the "mode" of the individual trees, providing superior accuracy over single-tree models.
- **Probabilistic Scoring:** Instead of a binary "Yes/No," the system utilizes `predict_proba`. This provides a percentage-based confidence level, which is essential for forensic investigation.
- **Resistance to Overfitting:** By using bagging and feature randomness, the model stays robust even when scammers change their tactics slightly.
- **Performance Metrics:** The model focuses on the "F1-Score"—balancing the need to catch thieves (Recall) with the need to avoid annoying real customers (Precision).



Technical Breakdown: The Analysis API (app.py)

- **Request Handling:** The `@app.route('/analyze')` function manages the POST request, ensuring the file is valid and readable.
- **Data Orchestration:** Converting the raw CSV buffer into a structured Pandas DataFrame for high-speed feature extraction.
- **Model Inference:**

```
# Load and Preprocess
df = pd.read_csv(file)
features = df.drop(columns=['id'])

# Generate Probabilities
probabilities = model.predict_proba(features)[: , 1]
predictions = model.predict(features)
```

Technical Breakdown: Risk & Summary Logic

- **Human-Centric Categorization:** The backend translates raw percentages into a "Traffic Light" risk system:
 - **HIGH (>80%):** Immediate red flag; likely unauthorized access or stolen credentials.
 - **MEDIUM (50-80%):** Anomalous behavior; requires secondary authentication or manual review.
 - **LOW (<50%):** Verified pattern; standard transaction.

- **Strategic Summary Logic:**

```
# Calculating executive stats
summary = {
    "total": len(df),
    "fraud": int(sum(predictions)),
    "normal": int(len(df) - sum(predictions))
}
```

- **JSON Response:** The data is sent as a structured object, making it easy for any UI to display results dynamically.

Cybersecurity, Privacy & DPDP Compliance

- **Data Minimization:** Adhering to the principle that only the minimum data required for a specific purpose (fraud detection) should be processed.
- **Volatile Processing:** Data is handled in RAM and not permanently written to the server's disk, significantly reducing the "Attack Surface" for data breaches.
- **IT Act & DPDP 2023:** The system is designed to provide "Reasonable Security" under Section 43A of the IT Act while respecting the consent and privacy mandates of the new Digital Personal Data Protection Act.
- **Input Sanitization:** Strict validation of CSV structures prevents common exploits like CSV Injection or Malicious File Uploads.



Project Impact, Challenges & Future Horizons

- **Operational Impact:** Drastically reduces the time-to-detection, allowing financial institutions to block fraudulent funds before they are withdrawn.
- **Key Challenges:**
 - **Model Drift:** The ongoing battle of retraining the AI as new scam patterns emerge.
 - **Latency:** Maintaining millisecond response times as the transaction volume scales.
- **The Future:**
 - **Explainable AI (XAI):** Integrating SHAP values to explain *why* the AI flagged a transaction.
 - **Real-time Integration:** Moving toward a Live-API model for instant, "at-the-swipe" protection.

THANK YOU!

